ClimateCraft – Glacier Segmentation & Area Extraction

Assigned Stage: Sentinel-2 imagery → Glacier segmentation → Final binary mask → Glacier area

This document defines one team member's complete stage of the project. The responsibility ends after producing validated glacier masks and the corresponding glacier-area dataset. It does not include climate-data collection, forecasting, XGBoost, backend development, or dashboard development.

1. Objective

The objective is to identify the glacier-covered region in each Sentinel-2 image and convert it into a binary segmentation mask. The mask is then used to calculate the observed glacier ice-covered area for each year.

Final concept: Satellite image → NDSI → initial snow/ice mask → GLIMS spatial reference → refined glacier mask → validation → pixel count → glacier area.

2. Input Data

The assigned person receives the scientific Sentinel-2 GeoTIFF files already collected for Samudra Tapu Glacier.

Input	Purpose
B2 – Blue	Spectral information; useful for preprocessing and analysis.
B3 – Green	Used with SWIR for NDSI calculation.
B4 – Red	Used for RGB visualization.
B8 – NIR	Useful for distinguishing vegetation/land and spectral analysis.
B11 – SWIR	Used with Green band for NDSI.
B12 – SWIR	Additional SWIR information for refinement/analysis.

Expected files are named approximately: Samudra_Tapu_2018_Scientific.tif through Samudra_Tapu_2026_Scientific.tif. If a year has no valid image, record it rather than inventing a result.

3. Step 1 – Read and Preprocess the Image

Load the six bands from each scientific TIFF. Check that the bands have the same spatial extent and are correctly aligned. Handle NoData or invalid pixels before calculating indices. B11/B12 are natively 20 m while the exported product may be on a 10 m grid, so the preprocessing should consistently handle this resampling.

4. Step 2 – Calculate NDSI

The Normalized Difference Snow Index (NDSI) is used to identify pixels that have snow/ice-like spectral behavior.

```
NDSI = (B3 – B11) / (B3 + B11)
```

Calculate this for every valid pixel. The result is a continuous NDSI raster. Higher positive values generally indicate snow/ice-like surfaces.

5. Step 3 – Create the Initial Snow/Ice Mask

Choose and document an NDSI threshold. A common starting point is 0.4, but the threshold should be tested against the actual imagery rather than treated as universally correct.

For each pixel:

```
if NDSI > threshold → mask = 1
if NDSI ≤ threshold → mask = 0
```

At this point the result is an initial snow/ice candidate mask. It may include seasonal snow outside the glacier.

6. Step 4 – Use the GLIMS Glacier Boundary

Use the GLIMS glacier boundary as the geographic reference for Samudra Tapu Glacier. The purpose is to restrict the candidate snow/ice pixels to the known glacier region and reduce obvious detections of surrounding seasonal snow.

Important: The GLIMS boundary is a reference, not an assumption that the historical boundary is exactly the glacier's current boundary. The final segmentation must still be visually checked against the satellite image.

```
Conceptually: NDSI candidate mask + GLIMS spatial reference → refined glacier mask.
```

7. Step 5 – Refine the Segmentation

Inspect the RGB image and the mask together. Remove obvious false detections such as isolated noise, surrounding seasonal snow, clouds/cloud shadows when identifiable, and clearly non-glacier surfaces. Preserve connected glacier regions that are supported by the satellite image.

The purpose of this stage is not to make the mask visually perfect by arbitrary editing. Every refinement rule should be reproducible and documented.

8. Step 6 – Final Binary Glacier Mask

The final output for every processed year must be a binary raster:

Pixel value	Meaning
0	Non-glacier
1	Glacier / glacier-covered pixel

```
Example mask structure:
0 0 0 1 1
0 1 1 1 1
0 1 1 1 0
0 0 1 0 0
```

9. Step 7 – Validate the Mask

For every year, visually compare the original RGB image with the final mask. Confirm that the mask follows the visible glacier region and does not obviously include surrounding terrain or seasonal snow.

If a reliable ground-truth/reference mask is available, segmentation quality can additionally be quantified using IoU, Dice, precision, and recall. These metrics should not be reported as authoritative unless the reference labels are appropriate for the date and image.

10. Step 8 – Calculate Glacier Area

After the final mask is validated, count the pixels whose value is 1. For a 10 m × 10 m grid:

```
Pixel area = 10 × 10 = 100 m²

Glacier area (km²) = Number of glacier pixels × 100 / 1,000,000
```

Example: if the final mask contains 500,000 glacier pixels, the observed glacier-covered area is 50,000,000 m² = 50 km².

11. Required Deliverables

Deliverable	Required output
Final masks	glacier_mask_2018.tif ... glacier_mask_2026.tif (or only years with valid imagery)
Area dataset	CSV containing year and glacier_area_km2
NDSI outputs	NDSI rasters or reproducible intermediate outputs
Validation images	RGB + final mask comparison for each year
Method notes	Threshold, preprocessing, GLIMS use, refinement rules, and area calculation

12. Definition of Done

This stage is complete when the team member can provide a final binary glacier mask for every usable year, demonstrate that each mask was checked against the corresponding satellite image, and provide a glacier-area value for each validated mask.

Her work ends here. The next project stage can use the resulting historical glacier-area dataset. Climate-data collection and future prediction are separate team responsibilities.

One-line task to give the teammate:
“Take the Sentinel-2 scientific TIFFs, generate and refine the glacier segmentation using NDSI and the GLIMS reference boundary, validate the final 0/1 glacier masks, and calculate the observed glacier area for each year.”